



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



How AI is quietly rewiring Purdue Model, forcing industrial defenders to rethink trust across operational environments

Threat Intelligence Report

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1. Executive Summary

Lost Boys Cyber assesses with moderate-to-high confidence that AI-enabled industrial workflows are accelerating a shift from location-based OT trust to identity-, data-flow-, and decision-authority-based trust. The 16 June 2026 Industrial Cyber feature does not identify a single named adversary campaign; rather, it documents an emerging risk pattern in which AI copilots, predictive-maintenance platforms, historians, industrial data lakes, edge inference nodes, cloud analytics, and remote operations tooling create new cross-zone trust relationships that the classic Purdue Model does not fully govern.

For defenders, the central issue is not that the Purdue Model is obsolete. The model remains useful for understanding hierarchy, safety-critical segmentation, and zone/conduit design. The issue is that AI workloads increasingly behave as active intermediaries across IT, OT, cloud, and edge environments. They ingest plant-floor telemetry, query historians, summarize alarms, enrich maintenance workflows, send recommendations to operators, and receive model or policy updates from enterprise or cloud platforms. These pathways can become attacker-relevant conduits if identity, authorization, logging, model governance, and human-in-the-loop controls are not engineered as first-class safeguards.

Field	Assessment
Report ID	LBC-TAP-2026-010
Report Family	Threat Actor Profile / Threat Landscape Profile
Primary Analytic Focus	AI-enabled cross-domain trust pathways in OT environments
Source Trigger	Industrial Cyber feature published 2026-06-16
Primary Risk	Misgoverned AI/edge/cloud integrations weaken Purdue assumptions and create lateral pathways between enterprise, cloud, historian, engineering, and OT layers
Most Exposed Environments	Manufacturing, energy, utilities, transportation, oil and gas, water/wastewater, pharmaceuticals, maritime, and other critical infrastructure operators using AI-assisted OT analytics
Confidence	Moderate-to-high for architectural risk trend; moderate for adversary exploitation maturity due to limited public case evidence in the source
Priority	High for organizations deploying AI workflows against OT data or connecting plant-floor telemetry to cloud/edge AI platforms

Lost Boys Cyber recommends treating AI systems in OT as privileged, semi-autonomous trust brokers rather than as passive analytics tools. Priority actions include mapping AI data flows, enforcing identity-aware access to historians and engineering assets, applying micro-segmentation to AI conduits, requiring explicit human validation before safety- or production-impacting changes, monitoring model-update and inference pathways, and incorporating AI failure and abuse cases into OT incident response playbooks.

2. Intelligence Requirement

This report answers the following intelligence requirement:

Requirement	Answer
Primary question	How does AI adoption change trust assumptions in Purdue-modeled OT environments, and what should defenders monitor or control?
Secondary question	What adversary opportunities emerge when AI copilots, edge inference, data lakes, historians, and predictive maintenance

	platforms bridge IT, OT, cloud, and edge networks?
Operational decision supported	Whether OT security teams should treat AI integrations as ordinary application deployments or as high-trust cross-zone conduits requiring dedicated governance, detection, and hunt coverage
Time horizon	Immediate architecture review for current AI/OT pilots; 6–18 month roadmap for durable identity, segmentation, logging, and governance controls

Analytic assumptions used in this report:

- The source article is primarily an expert-perspective and trend analysis piece, not an incident report.
- No named threat actor, malware family, or confirmed intrusion cluster is attributed by the source.
- The report family is retained as Threat Actor Profile because the defensive value is a profile of the adversary-relevant capability and opportunity space: AI-assisted abuse of cross-domain OT trust.
- Detections and hunts below are generic starting points that require environment-specific tuning against each operator's historian, EDR, firewall, identity, cloud, and OT telemetry.

3.1 Source summary

The primary source is Industrial Cyber's 16 June 2026 article, “How AI is quietly rewiring Purdue Model, forcing industrial defenders to rethink trust across operational environments.” The article argues that Purdue-style segmentation remains useful but is no longer sufficient when AI workflows routinely connect historians, engineering workstations, enterprise applications, cloud data lakes, predictive maintenance systems, and edge inference nodes.

The source cites or discusses several key data points and expert assessments:

Source element	Relevance	Lost Boys Cyber evaluation
Industrial Cyber feature	Primary reporting and expert commentary	Credible industry reporting; article is analytic rather than forensic
CISA / ASD ACSC joint AI-in-OT guidance	Confirms government concern around safe AI integration in OT	Strong corroboration that AI-in-OT risk is a recognized critical-infrastructure issue
IDC edge-computing spending projection	Supports the claim that edge compute and AI workloads are scaling materially	Useful context for adoption pressure; not direct evidence of compromise
Expert commentary from NVIDIA, WEF, Darktrace, and Palo Alto Networks	Provides practitioner views on Purdue evolution, zero trust, telemetry, and governance	Useful for defensive architecture framing; vendor perspectives may emphasize modernization themes
PwC digital operations / trust survey references	Indicates business appetite for AI/cloud adoption and OT staffing challenges	Useful adoption context; survey details should be validated before being used as quantitative risk scoring inputs

3.2 Corroborating source notes

CISA and the Australian Signals Directorate's Australian Cyber Security Centre published “Principles for the Secure Integration of Artificial Intelligence in Operational Technology” on 3 December 2025 with international partners. CISA describes AI in OT as capable of improving efficiency and decision-making while introducing risks that require careful management to preserve safety, security, and reliability. CISA's public summary identifies four steps: understand AI, assess AI use in OT, establish AI governance, and embed safety and security.

IDC reporting on worldwide edge-computing spending projects nearly \$261 billion in 2025 spending, growing to roughly \$380 billion by 2028, with AI workloads and AI-accelerated processors driving investment. Manufacturing and resources are reported as the second-largest vertical grouping, collectively representing about a quarter of worldwide spending. This supports the Industrial Cyber article's core premise that AI/edge adoption is not theoretical in industrial environments.

3.3 Confidence assessment

Judgment	Confidence	Rationale
AI/edge/cloud workflows are increasing cross-zone OT data movement	High	Supported by the source, CISA guidance, and edge-compute spending/adoption signals
Purdue-style segmentation remains necessary but insufficient by itself	High	Consistent across cited experts and established OT security practice; aligns with IEC 62443 zone/conduit logic and zero trust adaptation guidance
Misgoverned AI integrations can create adversary-relevant trust bridges	Moderate-to-high	Strong architectural logic; public intrusion evidence specific to AI-in-OT remains limited
Adversaries will use AI to accelerate OT vulnerability discovery, path analysis, and social/technical targeting	Moderate	Plausible and supported by expert statements, but source does not provide incident-level proof
Autonomous AI directly controlling safety-critical processes is a near-term widespread reality	Low-to-moderate	Source indicates most deployments remain supervisory or decision-support; direct closed-loop control is likely limited and higher-risk

4. Key Findings

ID	Finding	Defensive implication	Priority
KF-01	AI workflows are becoming cross-domain trust brokers in OT environments.	Treat AI copilots, predictive maintenance systems, data lake connectors, and edge inference nodes as privileged conduits, not passive analytics.	High
KF-02	The Purdue Model still helps organize industrial hierarchy but cannot prove trust by location alone.	Zone membership must be supplemented with identity, least privilege, continuous verification, and operational-context-aware policy.	High
KF-03	Historian, engineering workstation, cloud analytics, and third-party maintenance integrations are likely high-value abuse paths.	Build explicit data-flow inventories and alert on abnormal historian reads, engineering asset access, and cloud connector behavior.	High
KF-04	AI creates a determinism gap: probabilistic recommendations can influence deterministic control environments.	Separate insight from authority; require human validation and safe fallback before recommendations affect production or safety decisions.	High
KF-05	Model updates, prompts, connectors, service accounts, and API tokens expand the OT attack surface.	Monitor AI platform administration, model/package updates, secrets use, and new outbound connections from OT-	Medium-High

		adjacent systems.	
KF-06	OT staffing shortages increase the risk of unreviewed AI pathways and shadow integrations.	Require architecture review gates for AI pilots and maintain an approved registry of AI/OT use cases and data flows.	Medium-High
KF-07	Defender AI can improve OT visibility, anomaly detection, and vulnerability prioritization when governed.	Use AI to augment analysts but do not let AI outputs bypass engineering validation or change-control.	Medium

5. Threat Context

The adversary opportunity described by the source is best understood as a trust-boundary problem. Classic Purdue diagrams assume that security-relevant control points are largely aligned with levels and zones: enterprise systems at higher levels, operations management and historians in middle layers, and safety/control assets in lower levels. Modern AI deployments disrupt that clean mental model because data and influence can cross levels through application-layer workflows that appear legitimate.

Relevant AI-enabled industrial pathways include:

Pathway	Typical business justification	Security concern
Historian to cloud data lake	Central analytics, fleet benchmarking, compliance reporting	Bulk operational data exposure; cloud identity or token compromise may reveal process behavior
Predictive maintenance platform to plant telemetry	Equipment reliability and reduced downtime	Third-party or cloud platform becomes a dependency for interpreting asset health and maintenance decisions
AI copilot to historian / alarm data	Faster troubleshooting and operator support	Prompt, identity, or connector abuse may expose sensitive operations or shape operator decisions
Edge inference node at Level 1/2/3	Low-latency anomaly detection, vision inspection, robotics, local optimization	Edge node may bridge OT telemetry, model updates, vendor remote access, and enterprise monitoring
Engineering workstation assisted by AI	Faster ladder logic review, configuration analysis, documentation	AI-assisted workflow may access sensitive project files and recommend unsafe or manipulated changes
Remote operations or vendor service application	Centralized expertise and reduced site visits	Remote identity compromise can become an operationally meaningful pathway despite network segmentation

Adversaries do not need to “break the Purdue Model” if a legitimate AI workflow already bridges zones. They can target the identity, connector, API token, model registry, cloud workspace, update channel, or operator decision process that the AI workflow depends on. This makes conventional network-layer allow/deny rules necessary but incomplete.

Likely adversary behaviors include:

- Credential theft or token theft against AI/analytics service accounts used to query historians or data lakes.
- Abuse of approved data export paths to stage reconnaissance against industrial process behavior.
- Compromise of edge inference hosts that maintain dual trust with OT telemetry and cloud update infrastructure.
- Manipulation of AI prompts, retrieval sources, or model context to influence engineering or operator decisions.

- Use of AI to accelerate vulnerability discovery, exploit path analysis, phishing personalization, and OT-specific reconnaissance.
- Living-off-the-land movement through IT systems that already have approved conduits into OT support environments.

MITRE ATT&CK mapping

Tactic	Technique	Relevance to AI/Purdue trust-bridge scenario
Initial Access	Phishing (T1566)	AI-themed or vendor-themed lures may target engineers, maintenance staff, and administrators with access to AI/OT platforms.
Initial Access	Valid Accounts (T1078)	Compromised cloud, historian, VPN, vendor, or AI service accounts can make cross-zone activity appear legitimate.
Discovery	Network Service Discovery (T1046)	Attackers may enumerate historian APIs, AI connectors, edge services, and remote operations portals.
Discovery	Cloud Service Dashboard / Cloud Service Discovery (T1538 / T1526)	Cloud AI and data lake environments can reveal OT datasets, connectors, model registries, and access policies.
Collection	Data from Information Repositories (T1213)	Historians, data lakes, engineering repositories, and AI retrieval stores may contain high-value operational context.
Command and Control	Web Service (T1102)	Approved cloud/AI API traffic can blend into normal outbound paths from OT-adjacent systems.
Lateral Movement	Remote Services (T1021)	Compromised AI/maintenance support accounts may pivot through jump hosts, remote access platforms, or engineering workstations.
Defense Evasion	Masquerading (T1036)	Malicious tools may be named as AI agents, data collectors, maintenance scripts, or telemetry connectors.
Impact	Inhibit System Recovery (T1490) / Service Stop (T1489)	If attackers move from AI support systems into operations support infrastructure, they may disrupt telemetry, analytics, or response.
ICS Impact	Loss of View / Loss of Control / Manipulation of View	AI or historian manipulation can degrade operator confidence, distort process visibility, or influence unsafe decisions even without direct PLC control.

6. Detection and Hunting

Detection strategy should focus on the relationships AI systems create: identities, conduits, telemetry paths, update paths, administrative actions, and decision points. Because OT environments vary widely, the examples below are starting points for

Microsoft Defender / Sentinel-style KQL and Splunk-style SIEM data. Field names must be adapted to local EDR, identity, firewall, proxy, historian, and OT monitoring schemas.

6.1 Detection coverage plan

Coverage area	Telemetry source	High-value signals	Why it matters
AI service account authentication	Entra ID / IAM / VPN / PAM logs	New source IP, impossible travel, interactive login by service account, failed MFA, new token grant	Service accounts may have historian, data lake, or connector privileges.
Historian and OT data access	Historian audit logs, API gateway logs, proxy logs	Bulk reads, unusual tags, after-hours queries, new client apps, writes where only reads are expected	AI tools often require broad read access; abuse can support process reconnaissance.
Edge inference node behavior	EDR, firewall, OT sensor, syslog	New outbound destinations, model downloads, package installs, remote shell, new listening ports	Edge nodes can sit near lower Purdue levels while maintaining cloud connectivity.
AI platform administration	Cloud audit logs, model registry logs, CI/CD logs	New model deployment, connector creation, prompt/tool policy change, secret read, new webhook	Model or connector changes can alter decision support and data movement.
Engineering workstation AI use	EDR, proxy, DLP, code repository logs	Uploads of project files, ladder logic exports, unusual AI API use, scripts modifying control project files	Engineering context can affect production and safety if manipulated.
Remote maintenance and third-party access	VPN, ZTNA, PAM, vendor portal logs	Vendor account access outside windows, access to AI/OT connectors, new data export destinations	Third-party maintenance systems may become implicit trust bridges.

6.2 KQL: AI service account anomalous authentication

```

let lookback = 14d;
let baseline = 30d;
let AIAccounts = dynamic(["ai-ot-connector", "hist-ai-reader", "predictive-maintenance", "ot-copilot", "edge-inference"]);
let historical = SigninLogs
| where TimeGenerated between (ago(baseline) .. ago(lookback))
| where UserPrincipalName has_any (AIAccounts) or AppDisplayName has_any (AIAccounts)
| summarize KnownIPs=make_set(IPAddress), KnownApps=make_set(AppDisplayName) by UserPrincipalName;
SigninLogs
| where TimeGenerated > ago(lookback)
| where UserPrincipalName has_any (AIAccounts) or AppDisplayName has_any (AIAccounts)
| join kind=leftouter historical on UserPrincipalName
| extend NewIP = iff(isempty(KnownIPs) or not(IPAddress in (KnownIPs)), true, false)
| extend NewApp = iff(isempty(KnownApps) or not(AppDisplayName in (KnownApps)), true, false)
| where ResultType != 0 or NewIP or NewApp or ConditionalAccessStatus =~ "failure"
| project TimeGenerated, UserPrincipalName, AppDisplayName, IPAddress, Location, ResultType, ResultDescription, ConditionalAccessStatus, NewIP, NewApp
| order by TimeGenerated desc

```

Purpose: identify service-account or application-authentication drift for accounts that connect AI platforms to historians, cloud data lakes, or OT-adjacent systems.

6.3 KQL: edge inference node new outbound destinations

```
let EdgeNodes = DeviceInfo
| where DeviceName has_any ("edge", "infer", "vision", "ot-ai", "maintenance-ai")
| summarize by DeviceName;
let baseline = DeviceNetworkEvents
| where Timestamp between (ago(30d) .. ago(7d))
| join kind=inner EdgeNodes on DeviceName
| summarize KnownRemote=make_set(RemoteUrl), KnownIPs=make_set(RemoteIP) by DeviceName;
DeviceNetworkEvents
| where Timestamp > ago(7d)
| join kind=inner EdgeNodes on DeviceName
| join kind=leftouter baseline on DeviceName
| extend remote = coalesce(RemoteUrl, RemoteIP)
| where not(remote in (KnownRemote)) and not(RemoteIP in (KnownIPs))
| where RemotePort in (22, 80, 443, 445, 502, 1883, 4840, 3389, 5985, 5986) or RemoteUrl has_any
("blob", "s3", "model", "registry", "openai", "api", "huggingface")
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, RemoteUrl,
RemoteIP, RemotePort, Protocol, ActionType
| order by Timestamp desc
```

Purpose: surface new cloud, model-registry, remote-management, or industrial-protocol communications from OT-adjacent AI/edge hosts.

6.4 KQL: historian bulk access and unusual tag collection

```
// Adapt table and field names to the deployed historian or API gateway logs.
let Window = 1h;
HistorianAudit_CL
| where TimeGenerated > ago(7d)
| summarize
    TagCount=dcount(TagName_s),
    ReadEvents=countif(Operation_s =~ "read"),
    WriteEvents=countif(Operation_s =~ "write"),
    UniqueAssets=dcount(AssetId_s),
    SampleTags=make_set(TagName_s, 10)
    by bin(TimeGenerated, Window), User_s, ClientHost_s, Application_s
| where TagCount > 500 or WriteEvents > 0 or UniqueAssets > 50
| project TimeGenerated, User_s, ClientHost_s, Application_s, TagCount, ReadEvents, WriteEvents,
UniqueAssets, SampleTags
| order by TimeGenerated desc
```

Purpose: detect bulk historian reads, unexpected writes, or broad tag access by AI connectors, cloud export jobs, or compromised accounts.

6.5 Splunk: AI/OT connector administrative changes

```
index=cloud_audit OR index=ai_platform OR index=ot_admin
(action="CreateConnector" OR action="UpdateConnector" OR action="DeployModel" OR action="RotateSecret"
OR action="CreateWebhook" OR action="UpdatePromptPolicy" OR action="GrantRole")
| eval actor=coalesce(user, actor, principal, src_user)
| eval target=coalesce(resource, connector_name, model_name, app, service_principal)
| stats count min(_time) as first_seen max(_time) as last_seen values(action) as actions values(src_ip)
as src_ips values(target) as targets by actor
| where count > 2 OR mvcount(src_ips) > 1 OR match(mvjoin(actions, ","), "DeployModel|GrantRole|
RotateSecret")
| convert ctime(first_seen) ctime(last_seen)
| sort - last_seen
```

Purpose: identify concentrated or unusual administrative activity that could change how an AI workflow accesses OT data or influences decisions.

6.6 Sigma: suspicious AI connector touching engineering artifacts

```
title: AI or Analytics Connector Accessing Engineering Project Artifacts
id: 5a3a8e61-65db-45d4-9e5d-aiot-purdue-001
status: experimental
description: Detects AI, analytics, or predictive-maintenance processes accessing engineering
```

```

workstation project files that may contain PLC logic, HMI projects, or control-system configuration.
author: Lost Boys Cyber
date: 2026-06-16
logsource:
  category: file_event
  product: windows
detection:
  selection_process:
    Image|contains:
      - '\\ai'
      - '\\copilot'
      - '\\analytics'
      - '\\predictive'
      - '\\connector'
      - '\\python.exe'
      - '\\node.exe'
  selection_files:
    TargetFilename|contains:
      - '.ACD'
      - '.L5X'
      - '.L5K'
      - '.RSS'
      - '.MER'
      - '.APA'
      - '.ZAP'
      - '.S7P'
      - '.TIA'
      - '.VSD'
      - '.PVA'
      - '.HMI'
  filter_known_paths:
    Image|contains:
      - '\\Program Files\\ApprovedVendor\\'
      - '\\Program Files\\HistorianVendor\\'
  condition: selection_process and selection_files and not filter_known_paths
fields:
  - UtcTime
  - Image
  - User
  - TargetFilename
  - ComputerName
  - Hashes
falsepositives:
  - Approved engineering backup tools
  - Approved source-control or DLP indexing jobs
level: medium

```

Purpose: detect AI/analytics tooling touching engineering project artifacts outside approved workflows.

6.7 Threat-hunting hypotheses

The report does not identify confirmed adversary infrastructure or file-based indicators of compromise. For this reason, the IOC posture is deliberately conservative: treat the names below as local tuning seeds and exposure indicators, not global malicious IOCs.

Indicator / exposure artifact	Type	Defensive use	Confidence
AI/OT connector service accounts such as `ai-ot-connector`, `hist-ai-reader`, `predictive-maintenance`, `ot-copilot`, and `edge-inference`	Local identity pattern	Seed identity detections for service-account drift, interactive login, new token grants, and unexpected source IPs.	Low as universal IOC; medium as local exposure marker
Edge AI hosts named with `edge`, `infer`, `vision`, `ot-ai`,	Local asset pattern	Seed EDR/network hunts for new outbound destinations,	Low as universal IOC; medium

or `maintenance-ai`		remote shell, package installs, model downloads, or industrial protocol use.	as local exposure marker
Cloud/model registry/API destinations reached from OT-adjacent hosts	Network exposure pattern	Investigate new model-update, storage, API, or webhook destinations from Level 2/3 systems and edge nodes.	Medium when newly observed
Historian bulk reads, broad tag queries, or unexpected writes by AI workloads	Behavior indicator	Triage data-exfiltration, reconnaissance, unsafe automation, or connector misconfiguration.	Medium-to-high when anomalous to baseline
Hypothesis	Hunt action	Expected benign explanation	Escalation condition
A compromised AI connector is performing reconnaissance through historian data.	Compare historian tag breadth, time-of-day, and client hosts against 30-day baseline.	Scheduled analytics export or model training job.	New account, new host, unusual tags, or access outside change window.
An edge inference node is acting as an unintended bridge between cloud and lower-level OT networks.	Correlate edge node outbound cloud traffic with inbound or lateral OT protocol traffic.	Approved model update followed by normal inference.	New remote destination plus SMB/RDP/SSH/Modbus/OPC activity.
An AI assistant is exposing engineering artifacts to unauthorized services.	Review proxy/DLP/EDR logs for engineering file extensions uploaded to AI APIs or unapproved SaaS.	Approved documentation workflow with redaction.	Upload of PLC/HMI project files, configuration exports, or network diagrams.
A model or prompt-policy change altered operator decision support.	Review model registry, prompt policy, connector, and deployment logs around anomalous recommendations.	Planned model update with validation evidence.	Change lacks approval, validation, or rollback plan.
Adversaries are using AI/OT terminology to hide tooling.	Search endpoints for newly created services, scheduled tasks, containers, or scripts named like analytics/AI connectors.	New approved deployment.	Unsigned binary, unusual parent process, abnormal network destination.

7. Recommendations

Priority	Recommendation	Owner	Target state
1	Build an AI/OT data-flow inventory covering historians, data lakes, copilots, predictive maintenance tools, edge nodes, vendors, and cloud services.	OT Security / Architecture	Every AI workflow has documented data sources, identities, conduits, privileges, and business owner.
2	Treat AI connectors and service accounts as high-risk privileged identities.	IAM / OT Security	MFA or workload identity controls, least privilege, no shared secrets, token rotation, PAM where feasible, and alerting on interactive use.
3	Enforce Purdue-plus	Network / OT Engineering	AI systems must earn access by

	segmentation: zones and conduits plus identity-aware policy and continuous verification.		identity and purpose, not by network location alone.
4	Separate AI insight from operational authority.	Operations / Safety / Engineering	AI recommendations cannot directly change safety-critical or production-impacting controls without validated human approval and rollback.
5	Monitor model, prompt, connector, and data-pipeline changes.	SOC / Cloud Security	AI platform changes are logged, reviewed, and integrated into incident response.
6	Add AI failure and abuse cases to OT incident response playbooks.	IR Lead / OT Security	Playbooks cover model drift, poisoned retrieval data, connector compromise, cloud outage, unsafe recommendation, and data leakage.
7	Require architecture review before new AI/OT pilots move into production.	Governance / Risk	Review includes data classification, safety impact, network path, identity model, logging, kill switch, vendor access, and fallback mode.
8	Use defender AI cautiously for asset discovery, anomaly triage, and vulnerability prioritization.	SOC / Detection Engineering	AI augments analysts but does not automatically authorize remediation or control changes.

Implementation guidance:

- Start with the highest-risk conduits: historian-to-cloud export, edge inference with cloud update capability, AI copilots used by engineers/operators, and third-party predictive maintenance access.
- Add explicit labels in CMDB/asset inventory for AI-connected OT assets and service accounts.
- Require change tickets for model deployments and connector policy updates in the same way OT teams require change control for engineering workstations and remote access.
- Build tabletop scenarios where an AI recommendation is wrong, manipulated, or based on compromised data, then test whether operators can identify, challenge, and safely override the output.
- Align controls to CISA's four principles for AI in OT: understand AI, assess AI use in OT, establish AI governance, and embed safety and security.

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